

DTSA 5705

Stability and Capability in Quality Improvement

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	ML Engineer (74.9 / 100)	64.8 / 100
SPECIALIZATION Statistical Quality Improvement		

65

CURRENT AVG SCORE
across 6 job roles

96

POTENTIAL SCORE
with recommended additions

+38

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Students will learn to analyze data in terms of process stability and statistical control and why having a stable process is imperative prior to performing statistical hypothesis testing. Students will create statistical process control charts for both continuous and discrete data using R software. Students will analyze data sets for statistical control using control rules based on probability. Additionally, students will learn how to assess a process with respect to how capable it is of meeting specifications, either internal or external, and make decisions about process improvement.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
ML Engineer	74.9	99.2	+24.3	MEDIUM
Data Scientist	74.0	100.0	+26.0	MEDIUM
AI Engineer	69.1	100.0	+30.9	HIGH
Machine Learning Engineer	63.4	99.5	+36.1	HIGH
Applied Scientist	60.6	98.3	+37.7	HIGH
MLOps Engineer	47.0	76.1	+29.1	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 4 of 6 roles

systems

LLMs & Generative AI 1 of 6 roles

storage

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: LLMs & Generative AI** — would broaden appeal to one additional role. Candidate terms: storage.

3 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

ML Engineer

74.9 → 99.2 (+24.3 pts)

learning machine systems models engineering build problems impact

Data Scientist

74.0 → 100.0 (+26.0 pts)

presentation mitigation build engineering problems

AI Engineer

69.1 → 100.0 (+30.9 pts)

engineering models learning build technical systems problems technology

Machine Learning Engineer

63.4 → 99.5 (+36.1 pts)

engineering learning tools exploratory business design science models

Applied Scientist

60.6 → 98.3 (+37.7 pts)

computational storage problems models technical learning systems engineering

MLOps Engineer

47.0 → 76.1 (+29.1 pts)

detection systems learning machine image engineering tools models

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥ 200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.